Thermodynamics

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Contents

1 Energy and Enthalpy

1.1 Potential and Kinetic Energy

Core Concept 1.1: Thermodynamics

Thermodynamics examines the **energy changes** that accompany chemical reactions and phase transitions, enabling us to predict both whether a reaction will proceed **spontaneously** and **how much energy it will release or absorb**.

In chemistry, energy is broadly classified into two fundamental categories.

Core Concept 1.2: Kinetic Energy

Kinetic Energy (KE) refers to the energy associated with the motion of a molecule and is broadly categorize into **vibrational**, **rotational**, and **translational** motion.

Core Concept 1.3: Potential Energy

Potential Energy (PE) refers to the energy stored within intra- and intermolecular interactions. These interactions are governed by electrostatic attractions between opposing charges (e^- *and protons*) within a system, a principle formally described by **Coulomb's Law**.

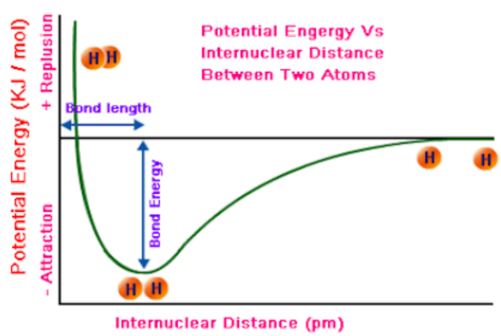

Figure 1: Bond potential energy diagram

As shown in Figure 1, intramolecular bonds arrange themselves at internuclear distances that **minimize PE**. When the internuclear distance is **infinitely large**, PE approaches zero, as the force between atoms is negligible.

As atoms approach one another, they reach an equilibrium point at which repulsive (*like-charge*) and attractive (*opposing-charge*) forces balance. The PE at this equilibrium is defined as the **bond energy**, and its formation is accompanied by a release of energy—analogous to two magnets snapping together. Conversely, compressing atoms beyond this equilibrium causes PE to rise, as repulsive forces begin to dominate—analogous to two rubber balls being forced together. *Bond energy is more rigorously characterized by the thermodynamic quantity* ***enthalpy****, discussed further in Core Concept 1.8.*

These observations give rise to two fundamental principles.

- **Breaking** a bond requires an **input of energy**; the change in PE from the bonded to unbonded state is therefore positive.
- **Forming** a bond releases an **output of energy**; the change in PE from the unbonded to bonded state is therefore negative.

Core Concept 1.4: The First Law of Thermodynamics

Within a **closed system**, the total energy is defined as the sum of its kinetic and potential energies:

$$E_{total} = KE + PE$$

This relationship reflects the **First Law of Thermodynamics**, which states that energy is neither created nor destroyed, but interconverted between KE and PE. When **PE decreases**—for instance, as two atoms form a bond and settle into a lower-energy configuration—the system gains a corresponding amount of KE. This conversion underlies **exothermic** reactions (*Core Concept 1.5*), in which the stabilization of chemical bonds manifests as increased molecular motion and the release of heat.

Conversely, when **KE decreases**—as when a molecule decelerates against an attractive intermolecular force—that energy is not lost, but transferred into PE as the molecule ascends a potential energy well shown in Figure 1

1.2 Heat

Core Concept 1.5: Endothermicity & Exothermicity

Chemical reactions can be classified based on the direction of heat flow between the system and its surroundings. In an **exothermic reaction**, energy is **released** into the surroundings—the products are at a lower energy state than the reactants, and the excess energy is dissipated as thermal energy. *Take the combustion of gasoline as an example.*

Conversely, in an **endothermic reaction**, energy is **absorbed** from the surroundings in order to drive the reaction forward—the products are at a higher energy state than the reactants, necessitating a net input of thermal energy. *A familiar example is the dissolution of ammonium nitrate in water, the principle underlying instant cold packs: as the reaction proceeds, it draws heat from the surrounding environment, producing a pronounced drop in temperature.*

Core Concept 1.6: Heat Transfer & Calorimetry

Heat transfer occurs when two systems at differing temperatures come into contact, whereupon thermal energy flows from the system of higher temperature to that of lower temperature.

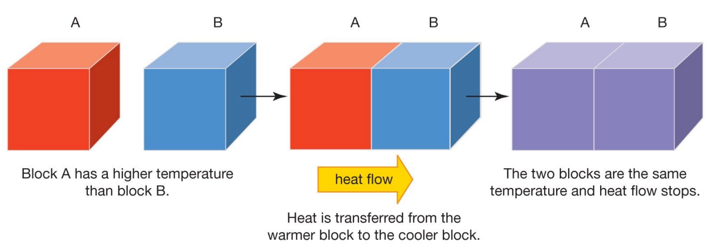

Figure 2: Heat transfer

This behavior reflects the underlying molecular basis of temperature: **hot** substances possess particles with greater average KE—vibrating and moving more rapidly—whereas **cold** substances possess particles with comparatively lower KE. **Temperature**, therefore, is more precisely defined as a measure of the **average kinetic energy** of the particles within a system.

The quantity of heat transferred within a system is given by:

$$q = m \times C \times \Delta T$$

where q denotes the heat transferred, m the mass of the system, C the **specific heat capacity**, and ΔT the change in temperature. **Specific heat capacity** is defined as the quantity of energy required to raise the temperature of 1 g of a substance by 1 K. For water, this value is 4.184 $J\,g^{-1}K^{-1}$, meaning 4.184 joules of energy are required to raise the temperature of 1 g of water by 1 K.

The experimental discipline concerned with measuring the heat exchanged during chemical reactions, physical changes, or thermal interactions between objects is known as **calorimetry**, and constitutes a practical application of the **First Law of Thermodynamics**. Consider the following worked examples.

(a) **Determine the heat required to raise the temperature of 43g of water by 45K.**

$$q = (43\,g)(4.184\,J\,g^{-1}K^{-1})(45\,K) = 8096.04\,J$$

While the joule is the standard unit of energy in chemistry, the **calorie** is another commonly encountered unit, defined as 4.184 J. Notably, this equivalence renders the specific heat capacity of water exactly $1\,cal\,g^{-1}K^{-1}$. It should be noted that the dietary Calorie (*Cal*), as reported on nutritional labels, corresponds to 1 kilocalorie (*kcal*), or 1000 calories.

(b) **A 560 g sample of Au at $100^{\circ}C$ is placed into 560 mL of water at $20^{\circ}C$. Determine the final equilibrium temperature of the system. Given: $C_{H_2O} = 4.184\,J\,g^{-1}K^{-1}$, $C_{Au} = 0.13\,J\,g^{-1}K^{-1}$.**

By the conservation of energy, the total heat change within a closed system must equal zero:

$$q_{lost} + q_{gained} = 0 \quad \Longrightarrow \quad q_{gained} = -q_{lost}$$

Since heat flows from the Au to the water, the Au loses energy while the water gains energy, and both reach a common final temperature T_{final}. Substituting into the calorimetry equation:

$$(560\,g)(4.184\,J\,g^{-1}K^{-1})(T_{final} - 20) = -(560\,g)(0.13\,J\,g^{-1}K^{-1})(T_{final} - 100)$$

Solving for T_{final}:

$$T_{final} = 22.41^{\circ}C$$

Core Concept 1.7: Phase Transitions

Phase transitions refer to the physical transformation of a substance from one state of matter to another, driven by changes in thermal energy. The heat required to transition water from a solid to a liquid state is termed the **heat of fusion**, while that required to transition water from a liquid to a gaseous state is termed the **heat of vaporization**.

The sign of each quantity is determined by the direction of the transition: positive for melting and vaporization (endothermic), and negative for freezing and condensation (exothermic).

As illustrated in Figure 3, temperature remains constant during phase transitions. During vaporization, for instance, the energy supplied does not increase the average KE of the molecules (*which temperature measures*), but is instead directed toward increasing the **PE of the IMFs**. At 100°C, water molecules in the liquid phase possess the same average KE as those in the gas phase; however, they remain confined to the liquid state by the deep PE wells created by hydrogen bonding interactions. The heat of vaporization, therefore, represents **the energy required to overcome these IMFs entirely**, liberating the molecules from their neighbors and enabling the transition to the gas phase. *The same reasoning applies analogously to the heat of fusion.*

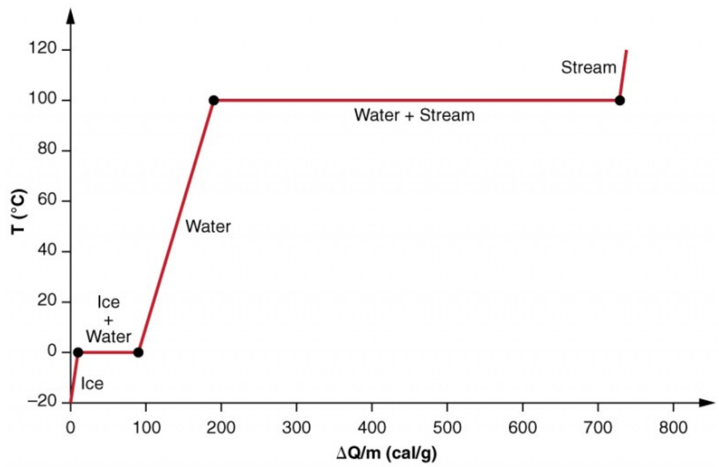

Figure 3: Heating curve of water

It is also notable that the heat of vaporization is substantially greater in magnitude than the heat of fusion. This disparity arises because **fusion** requires only sufficient energy to **disrupt the rigid, ordered crystal lattice** of ice—partially weakening intermolecular interactions without eliminating them—whereas **vaporization** necessitates the **complete dissociation** of all intermolecular interactions between water molecules. The energetic difference between these two processes is therefore considerable.

1.3 Enthalpy

Core Concept 1.8: Enthalpy

Enthalpy (H) is a thermodynamic state function representing the **total heat content** of a system at constant pressure. Formally, it is defined as the sum of the system's internal energy and the pressure-volume work required to establish its physical extent within the surrounding environment—*that is, the energy expended to displace the atmosphere and "make room" for the system*.

Although the absolute enthalpy of a system cannot be determined experimentally, the **change in enthalpy** (ΔH) during a chemical or physical process is directly measurable, making it one of the most practically significant quantities in thermochemistry.

To understand the molecular origin of ΔH, recall the potential energy diagram (*Figure 1*) introduced in Core Concept 1.3.

At the molecular level, all chemical reactions involve the breaking of bonds in the reactants and the formation of new bonds in the products. Since **bond formation releases energy** and **bond breaking requires energy**, the net enthalpy change of a reaction reflects the difference between these two contributions. When the energy released upon forming product bonds exceeds the energy required to break reactant bonds, the reaction is classified as **exothermic** ($\Delta H < 0$); conversely, when more energy is consumed than released, the reaction is **endothermic** ($\Delta H > 0$).

Core Concept 1.9: Enthalpy Example (*Combustion of Hydrogen*)

Take the combustion of hydrogen to produce water ($2\,H_2 + O_2 \longrightarrow 2\,H_2O$) as an illustrative example. The enthalpy change over the course of this reaction is depicted below.

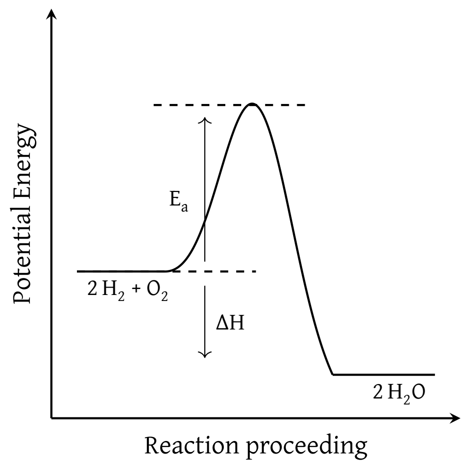

Figure 4: Enthalpy diagram of H_2 combustion

To build intuition, consider the reaction as it would appear in practice. Upon igniting a hydrogen-filled balloon, one observes an immediate release of heat and light—a clear indication that the reaction is exothermic. The original reactants (*H_2 within the balloon and O_2 from the surrounding air*) are converted into the product (H_2O) following an initial input of energy supplied by the ignition source. This initial energy input is termed the **activation energy** (E_a), which is the minimum energy required to **initiate the breakdown of reactant bonds** before product bonds can form.

The fundamental driving force underlying chemical reactivity is **the tendency of a system to minimise its potential energy**. Consequently, the free atoms produced upon bond dissociation—H, H, H, H, O, O—reorganise to form $2\,H_2O$, whose bonds exist at a lower potential energy state than those of the original reactants. The net difference in enthalpy between reactants and products is released as heat, manifesting as the observed explosion.

It is worth noting that a single spark does not supply sufficient activation energy to convert all H_2 and O_2 molecules simultaneously. Rather, the initial spark provides enough energy to initiate the formation of a single H_2O molecule. The energy released by this event then serves as the activation energy for neighbouring reactant molecules, which in turn release energy that drives further reactions. This cascade propagates rapidly throughout the balloon, accounting for the near-instantaneous nature of the explosion.

The enthalpy change of a reaction can also be determined quantitatively using average bond enthalpies. Representative values for several common bonds are given in Table 1.

Bond	H–H	O–O	H–O	C–C	C=C	C≡C	C–O	O=O
H_{average} (kJ/mol)	436	146	463	348	612	838	360	498

Table 1: Average bond enthalpies of common bonds

Applying these values to the combustion of hydrogen, the enthalpy change is calculated by summing the energies of bonds broken (*endothermic, positive sign*) and bonds formed (*exothermic, negative sign*):

- Break 2 mol of H–H → (2)(+436 kJ mol^{-1})

- Break 1 mol of O=O → (1)(+498 kJ mol^{-1})
- Form 4 mol of O–H → (4)(−463 kJ mol^{-1})

Thus $\Delta H = -482$ kJ mol^{-1} for the reaction. Finally, it is important to note that E_a does not represent the energy required to break all reactant bonds simultaneously. **E_a is only the energy needed to break key reactant bonds to the extent necessary for the reaction to proceed**. For instance, the activation energy for water formation is approximately 200 kJ mol^{-1} — far less than the 1370 kJ mol^{-1} that would be required for complete dissociation of all reactant bonds.

Core Concept 1.10: Enthalpy Example (*Autoionization of Water*)

Consider the autoionization of water as another example.

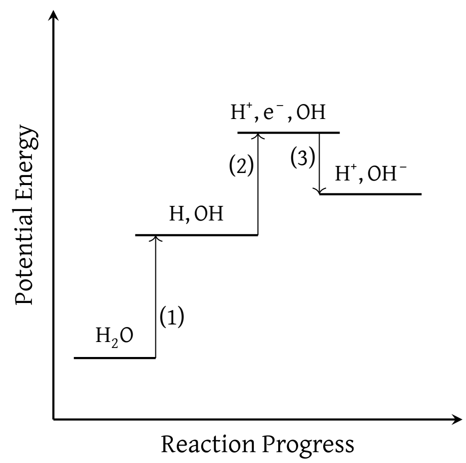

Figure 5: Enthalpy diagram for water autoionization

The diagram above in Figure 5 shows a net rise in PE, indicating that the overall reaction is endothermic ($\Delta H > 0$). The process can be decomposed into three distinct steps, each with an associated enthalpy contribution:

(1) Break 1 mol of O–H → (1)(+467 kJ mol^{-1})

(2) Remove e^- from 1 mol of H (*ionisation energy*) → (1)(+1312 kJ mol^{-1})

(3) Add e^- to 1 mol of O (*electron affinity*) → (1)(−180 kJ mol^{-1})

Thus $\Delta H = +1599$ kJ mol^{-1} for the reaction.

Core Concept 1.11: Hess's Law

When two or more chemical equations are combined to yield an overall reaction, the enthalpy of that reaction is equal to the sum of the enthalpies of the constituent steps. This principle is known as **Hess's Law**, and it follows directly from the fact that enthalpy is a **state function**—one whose value depends only on the initial and final states of the system, not on the path taken between them.

$$\Delta H_{\text{rxn}} = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

Conceptually, this expression corresponds to notionally **reversing the formation of the reactants** (*contributing a positive enthalpy term*) and then **carrying out the formation of the products** (*contributing a negative enthalpy term*), with the net enthalpy change reflecting the difference between the two.

Core Concept 1.12: Enthalpy Example (*Iron Smelting*)

Consider the reaction of iron oxide with carbon. The formation of iron oxide is an exothermic process, as given by the following equation.

$2\,Fe(s) + \frac{3}{2}O_2(g) \longrightarrow Fe_2O_3(s) \qquad \Delta H = -824\,\mathrm{kJ\,mol^{-1}}$

By Hess's Law, the reverse reaction—the decomposition of iron oxide—is endothermic with an equal and opposite enthalpy change. This is designated **Rxn A**.

(Rxn A) $Fe_2O_3(s) \longrightarrow 2\,Fe(s) + \frac{3}{2}O_2(g) \qquad \Delta H_A = +824\,\mathrm{kJ\,mol^{-1}}$

Next, consider the combustion of carbon, designated **Rxn B**.

(Rxn B) $C(s) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H_B = -394\,\mathrm{kJ\,mol^{-1}}$

Combining 2 equivalents of Rxn A with 3 equivalents of Rxn B yields the overall reduction of iron oxide by carbon, designated **Rxn C**.

(Rxn C) $2\,Fe_2O_3(s) + 3\,C(s) \longrightarrow 4\,Fe(s) + 3\,CO_2(g)$

Applying Hess's Law, the enthalpy of Rxn C is obtained by summing the scaled enthalpies of Rxn A and Rxn B.

$$\Delta H_C = 2\Delta H_A + 3\Delta H_B = +466\,\mathrm{kJ\,mol^{-1}}$$

This result indicates that deoxidising 2 moles of iron oxide requires $+466\,\mathrm{kJ\,mol^{-1}}$, or equivalently $+233\,\mathrm{kJ\,mol^{-1}}$ per mole—a substantially more energetically favourable outcome than the $+824\,\mathrm{kJ\,mol^{-1}}$ that would be required in the absence of carbon. This reaction forms the thermochemical basis of the iron smelting process, by which raw iron ore is reduced to pure iron.

Core Concept 1.13: Enthalpy Example (*Vaporization of Water*)

Consider the vaporization of water at 25 °C, in which liquid water is converted to its gaseous phase.

$H_2O(l) \longrightarrow H_2O(g) \qquad \Delta H_{\mathrm{rxn}} = ?$

This transformation can be decomposed into two intermediate steps. Rxn A represents the decomposition of liquid water into its constituent elements, and Rxn B represents the formation of gaseous water from those elements.

(Rxn A) $H_2O(l) \longrightarrow H_2(g) + \frac{1}{2}O_2(g) \qquad \Delta H_A = +286\,\mathrm{kJ\,mol^{-1}}$

(Rxn B) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g) \qquad \Delta H_B = -242\,\mathrm{kJ\,mol^{-1}}$

Applying Hess's Law, the enthalpy of vaporization is obtained by summing the enthalpies of the two steps.

$$\Delta H_{\mathrm{rxn}} = \Delta H_A + \Delta H_B = +44\,\mathrm{kJ\,mol^{-1}}$$

This result is consistent with the general formulation introduced previously, whereby the overall enthalpy change corresponds to **reversing the formation of the reactants** and **carrying out the formation of the products**.

$$\Delta H_{\mathrm{rxn}} = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

2 Entropy

2.1 Defining Entropy

Enthalpy alone is insufficient to predict whether a reaction will proceed spontaneously. The dissociation of ammonium chloride into its constituent ions, for instance, is endothermic yet proceeds to completion.

$$NH_4Cl \longrightarrow NH_4^+ + Cl^- \qquad \Delta H_{Rxn} = +14.8\, kJ\, mol^{-1}$$

Similarly, the spontaneous expansion of a gas into a vacuum occurs irreversibly despite $\Delta H = 0$, and thus cannot be accounted for by enthalpic considerations alone.

Core Concept 2.1: Entropy & The Second Law of Thermodynamics

There exists a natural tendency for energy and matter to disperse—in thermodynamic terms, systems tend to evolve towards states of higher **entropy**. This principle constitutes **the second law of thermodynamics**. Although introductory texts often characterise entropy as *disorder*, this description is imprecise. More rigorously, entropy is defined as a measure of the number of distinct microscopic configurations (**microstates**) consistent with a given observable state (**macrostate**).

Core Concept 2.2: Entropy (*Dice Analogy*)

Consider the sum obtained from rolling two dice. Across all 36 possible outcomes, the probability distribution is strongly concentrated around a sum of 7. In this context, the macrostate is defined by the sum, while the microstates are the individual dice combinations that yield it (*6–1, 5–2, 4–3, 3–4, 2–5, 1–6*). A low-entropy macrostate such as a sum of 2 has only a single accessible microstate, whereas the sum of 7 has six—making high-entropy macrostates statistically far more probable.

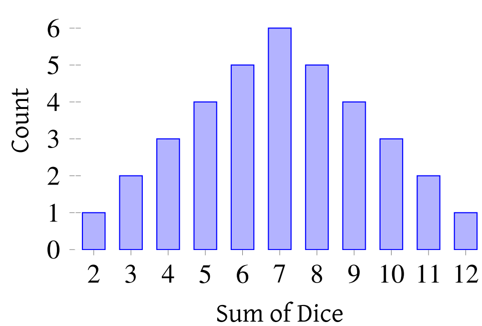

Figure 6: Dice roll probability distribution

Core Concept 2.3: Entropy (*Gas Molecule Traits*)

Consider a collection of gas molecules confined within a closed container. The spontaneous dispersion of the gas throughout the available volume is directly analogous to the dice example: uniform distribution across the container corresponds to the highest-entropy macrostate, as it is accessible through the greatest number of positional microstates. By contrast, the configuration in which all molecules are concentrated in a single corner of the container represents a low-entropy macrostate, accessible through comparatively few microstates. This is illustrated at a reduced scale in Figure 7.

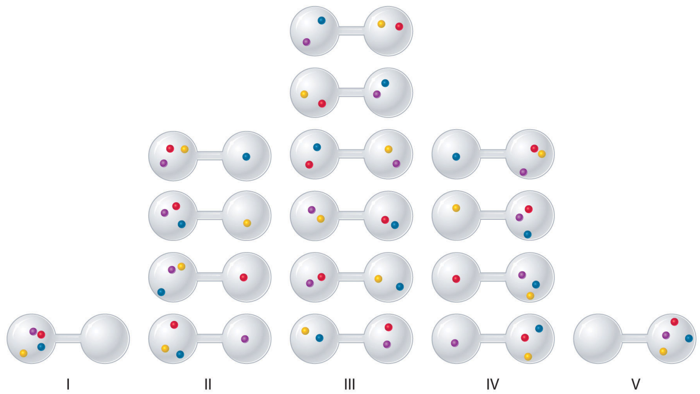

Figure 7: Gas molecule position microstates

The preceding discussion has focused exclusively on the positional coordinates of gas molecules. In reality, the number of accessible microstates for a given macrostate is further expanded by additional degrees of freedom, including rotational orientation, vibrational motion, translational velocity, and KE magnitude. Those dimensions associated with KE are, by extension, coupled to temperature. Just as positional microstates favour uniform spatial distribution, the microstates associated with KE favour its broad dispersal throughout the system—providing a statistical basis for the spontaneous transfer of heat from warmer to cooler regions.

It is important to note that when gas molecules are described as being at 25 °C, this refers to the *average* KE of the ensemble—not to all molecules simultaneously possessing the KE equivalent of 25 °C. At any given instant, individual molecules will deviate from this average, with some possessing slightly higher and others slightly lower KEs. Consequently, there exist far more microstates in which KEs are broadly distributed around the mean than those in which all molecules possess identical KEs.

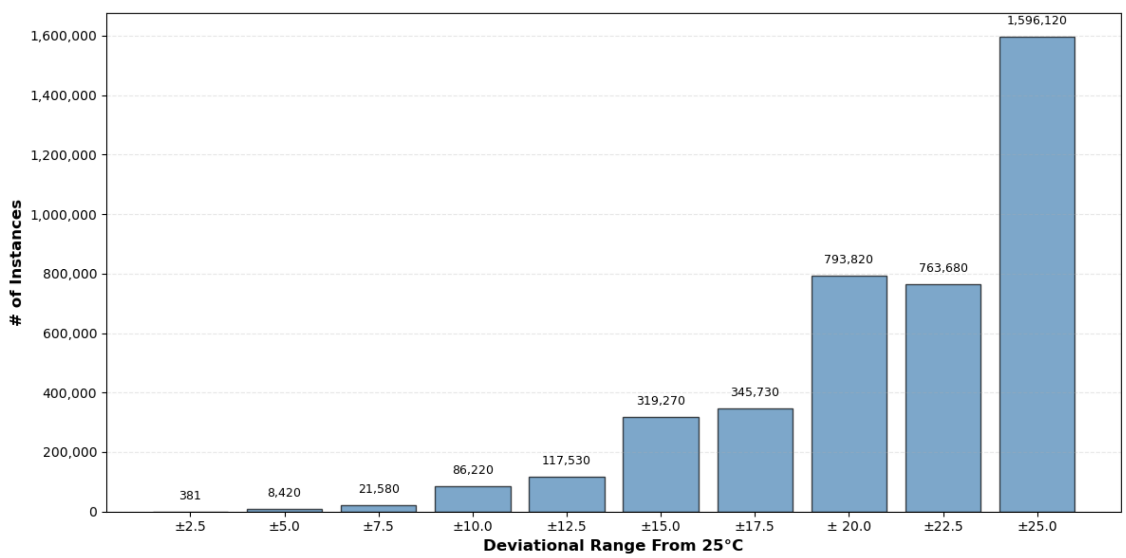

Figure 8: Gas kinetic energy distribution

Figure 8 plots the number of possible 5-integer permutations that sum to 125 (*and thus average to 25*), partitioned by the maximum permitted deviation from the mean. Even in this highly simplified model, the number of accessible microstates increases dramatically as the permitted spread in KE widens—vastly outnumbering the single microstate in which all values equal 25, such as the permutation (25, 25, 25, 25, 25). This statistical tendency for KE to be broadly and randomly distributed among gas molecules, while the ensemble average remains fixed, is the basis of the **Maxwell–Boltzmann Distribution** (Core Concept 3.2).

Core Concept 2.4: Boltzmann Equation

Entropy (S) is formally quantified through the **Boltzmann equation**:

$$S = k_B \ln(W)$$

where W denotes the number of energetically equivalent microstates accessible to the system, and k_B is the Boltzmann constant, defined as

$$k_B = \frac{R}{N_A} = \frac{8.314\, J\, K^{-1}\, mol^{-1}}{6.022 \times 10^{23}\, mol^{-1}} = 1.38 \times 10^{-23}\, J\, K^{-1}$$

where R is the ideal gas constant and N_A is Avogadro's number. Physically, k_B represents the amount of thermal energy contained per particle per degree of temperature.

As the preceding discussion illustrates, the number of accessible microstates grows exponentially with the number of particles in the system. It is for this reason that entropy is proportional to $\ln(W)$ rather than W directly—a relationship revisited in Core Concept 3.5.

In more general terms, entropy increases ($\Delta S > 0$) when

(a) Temperature increases.

(b) A reaction produces a net increase in the number of molecules.

(c) A phase transition occurs toward a less ordered phase (solid → liquid or liquid → gas).

In more nuanced cases, the guiding principle remains whether the process increases or decreases the number of accessible microstates.

2.2 Changes in Entropy

Core Concept 2.5: The Third Law of Thermodynamics

The **third law of thermodynamics** states that as the temperature of a system approaches absolute zero ($0\, K$), its entropy approaches a constant minimum value—defined as zero for a perfect, pure crystal. This is significant because it provides an absolute reference point from which the entropy of any system, molecule, or quantity of substance can be determined. This stands in contrast to enthalpy, for which no such absolute reference exists and only changes can be meaningfully quantified.

Core Concept 2.6: Entropy & Temperature

Entropy is expressed in units of **Joules per Kelvin ($J\, K^{-1}$)**. This unit is best understood by analogy: just as density ($g\, L^{-1}$) describes how much mass a substance occupies per unit volume, entropy describes the breadth of accessible microstates—quantified in units of energy—per unit temperature.

It is also important to note that **the entropy of a system increases with temperature, but at a diminishing rate.** To build intuition, consider supplying an equal quantity of energy to boiling water and to ice. Since boiling water already exists in a high-entropy state, a 1 K rise in temperature produces a proportionally smaller increase in entropy compared to the same temperature rise applied to ice, which begins in a low-entropy state. This relationship is expressed mathematically as

$$\frac{dS}{dT} = \frac{C_p}{T}$$

This expression states that the rate of entropy change with respect to temperature is proportional to the heat capacity of the system (C_p) and inversely proportional to the temperature (T). The inverse dependence on T reflects the diminishing effect described above. The proportionality to C_p follows from the fact that a higher heat capacity requires a greater energy input per kelvin of temperature increase—and consequently generates a greater expansion of accessible microstates. For instance, raising the temperature of water by 1 K requires substantially more energy than the same increase in copper; this large energy input into water produces a correspondingly greater increase in molecular disorder than the comparatively modest input required for copper.

Core Concept 2.7: Thought Experiment (*Increasing the Temperature of Water*)

Consider the heating of water from 25 °C to 50 °C, examined through the dual lenses of enthalpy and entropy.

From an enthalpic standpoint, this is a straightforward calorimetric calculation: ΔH is given by the product of the molar quantity of H_2O, its heat capacity C_{H_2O}, and the temperature change ΔT.

To express the same energy input in terms of entropy, two factors must be accounted for simultaneously: (1) the monotonic increase in entropy with temperature, and (2) the diminishing rate at which this increase occurs. These conditions are summarised mathematically as

$$\frac{dS}{dT} > 0, \quad \frac{d^2S}{dT^2} < 0$$

The entropy gained over this temperature interval corresponds to the area under the S versus T curve between the initial and final temperatures, reflecting the cumulative expansion of accessible microstates. The functional form of this curve resembles $f(x) = \ln(x)$, consistent with the constraints above.

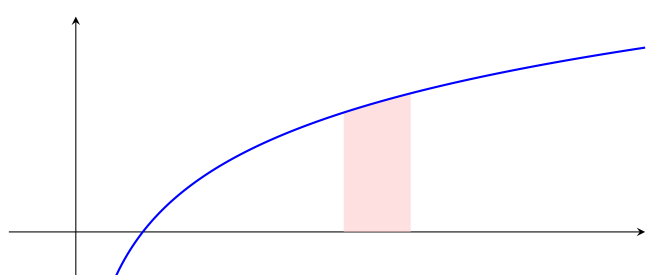

Figure 9: Plot of $\ln(x)$

On closer inspection, enthalpy can be understood as a macroscopic surrogate for the entropy change of the *surroundings*—that is, the thermal, temperature-dependent component of total entropy change. A complete thermodynamic description requires additionally accounting for the entropy change of the *system* itself, which captures configurational and positional contributions that enthalpy is blind to. It is the sum of these two contributions that ultimately determines spontaneity—a relationship formalized by Gibbs free energy (*Core Concept 3.2*).

3 Gibbs Free Energy

3.1 Gibbs Free Energy Foundations

Core Concept 3.1: Deriving the Gibbs Free Energy Equation

Returning to the **second law of thermodynamics**—which states that the total entropy of an isolated system increases over time—and treating the universe as an isolated system, this law can be expressed as the following.

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{environment} > 0$$

To illustrate, consider the melting of 10 moles of ice. Since temperature remains constant during a phase transition, the entropy change of the system can be calculated directly from the enthalpy of fusion without integration.

$$\Delta S_{system} = \frac{\Delta H_{fusion}}{T_{system}} = \frac{(6.02\,\mathrm{kJ\,mol^{-1}})(10\,\mathrm{mol})}{273\,\mathrm{K}} = +220\,\mathrm{J\,K^{-1}}$$

The temperature of the surroundings does technically decrease upon absorbing heat from the system, but this change is negligible given that the surroundings are modelled as a **thermal reservoir**—a body so large that its heat capacity is effectively infinite relative to the system, and its temperature remains constant at 298 K.

The corresponding entropy change of the surroundings is therefore

$$\Delta S_{surroundings} = \frac{-\Delta H_{fusion}}{T_{surroundings}} = \frac{-(6.02\,\mathrm{kJ\,mol^{-1}})(10\,\mathrm{mol})}{298\,\mathrm{K}} = -202\,\mathrm{J\,K^{-1}}$$

The total entropy change of the universe is then

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} = 220\,\mathrm{J\,K^{-1}} + (-202\,\mathrm{J\,K^{-1}}) = +18\,\mathrm{J\,K^{-1}}$$

This result reflects two competing contributions.

(1) The entropy of the system increases upon absorption of heat.

(2) The entropy of the surroundings decreases as heat is transferred from it—**but to a lesser degree**, since the same quantity of energy represents a smaller change at the higher temperature of the surroundings.

Algebraically manipulating the expression for $\Delta S_{universe}$ yields the **Gibbs free energy** equation. Substituting $\Delta S_{surroundings} = -\frac{\Delta H_{sys}}{T_{surr}}$:

$$\Delta S_{uni} = \Delta S_{surr} + \Delta S_{sys} > 0$$

$$\Delta S_{uni} = \frac{-\Delta H_{sys}}{T_{surr}} + \Delta S_{sys} > 0$$

$$T_{surr}\Delta S_{uni} = -\Delta H_{sys} + T_{surr}\Delta S_{sys} > 0$$

$$-T_{surr}\Delta S_{uni} = \Delta H_{sys} - T_{surr}\Delta S_{sys} < 0$$

Defining $\Delta G = -T_{surr}\Delta S_{uni}$, the condition for spontaneity becomes

$$\Delta G = \Delta H_{sys} - T_{surr}\Delta S_{sys} < 0$$

Core Concept 3.2: Gibbs Free Energy

Gibbs free energy (ΔG) provides a unified criterion for thermodynamic spontaneity, effectively quantifying the net tendency of the universe's entropy to increase. It represents a balance between two competing thermodynamic driving forces.

(1) The tendency to maximise the entropy of the universe

(2) The tendency to minimise the enthalpy of the system.

When both conditions are satisfied simultaneously, the reaction is unconditionally **spontaneous**; when neither is satisfied, the reaction is unconditionally **non-spontaneous**; when only one is satisfied, spontaneity is determined by the relative magnitudes of the two contributions.

ΔH°	ΔS°	T	ΔG°	Reaction Favorability
−	+	ANY	−	Product favored (→)
−	−	HIGH	+	Reactant favored (←)
−	−	LOW	−	Product favored (→)
+	+	HIGH	−	Product favored (→)
+	+	LOW	+	Reactant favored (←)
+	−	ANY	+	Reactant favored (←)

Table 2: Reaction favorability across different ΔH and ΔS values

ΔG_{rxn} quantifies the difference in free energy between products and reactants under standard-state conditions, and—like enthalpy and entropy—is a state function. It may be calculated either from the corresponding ΔH_{rxn} and ΔS_{rxn} values, or directly from tabulated standard Gibbs free energies of formation (ΔG_f).

$$\Delta G_{rxn} = \sum \Delta G_f(\text{products}) - \sum \Delta G_f(\text{reactants})$$

A useful way to interpret the Gibbs free energy equation is to treat $T\Delta S$ as a temperature-dependent scaling term that modulates the entropic contribution to spontaneity—amplifying or diminishing the influence of entropy depending on the thermal conditions of the system.

Core Concept 3.3: Gibbs Free Energy Example (*Barium Hydroxide & Ammonium Chloride Reaction*)

Consider the reaction between barium hydroxide octahydrate and ammonium chloride.

$$Ba(OH)_2 \cdot 8\,H_2O(s) + 2\,NH_4Cl(s) \longrightarrow BaCl_2(aq) + 2\,NH_3(aq) + 10\,H_2O(l)$$

Although the reaction is highly endothermic ($\Delta H = +62\,\mathrm{kJ\,mol^{-1}}$), inspection of the physical states of the reactants and products reveals a substantial increase in entropy ($\Delta S = +368\,\mathrm{J\,mol^{-1}\,K^{-1}}$), driven by the increase in the number of moles of molecules and the transition to more disordered phases. At 298 K, the magnitude of the entropic contribution $T\Delta S = +109.7\,\mathrm{kJ\,mol^{-1}}$ exceeds the enthalpic penalty, yielding $\Delta G < 0$ and confirming that the reaction proceeds spontaneously.

Core Concept 3.4: Thought Experiment (*Liquid Water at Absolute Zero*)

Consider the behaviour of water at a theoretical temperature of 0 K. It is well established that liquid water freezes at absolute zero, but the thermodynamic basis for this warrants examination. The relevant phase transition is represented by the following equation.

$$H_2O(l) \longrightarrow H_2O(s)$$

Freezing is exothermic and therefore enthalpically favourable. Although the transition from liquid to solid entails a decrease in entropy, this entropic penalty carries no thermodynamic weight at absolute zero—since $T\Delta S = 0$ when $T = 0\,K$, the entropic contribution to ΔG vanishes entirely. At absolute zero, Gibbs free energy reduces to enthalpy alone, and the exothermic nature of freezing makes the solid state unconditionally favoured. Note that while liquid water cannot physically exist at $0\,K$, this theoretical limiting case is precisely what the thermodynamic framework predicts and seeks to explain.

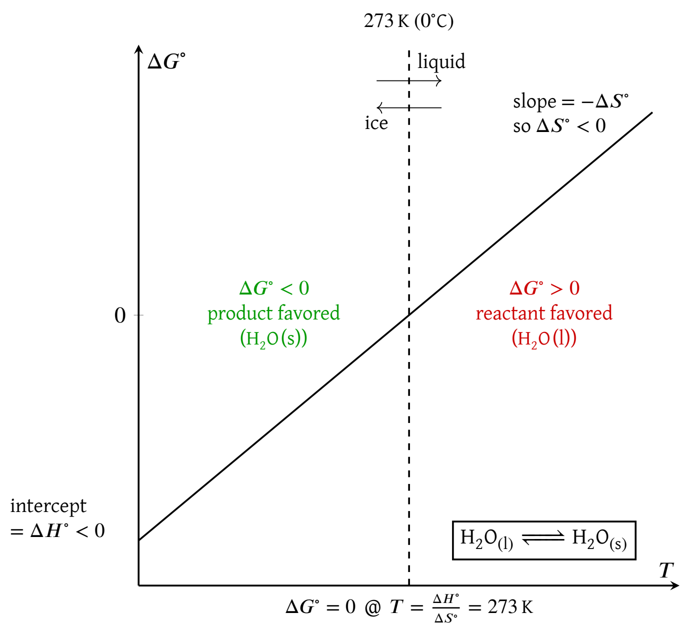

Figure 10: Gibbs free energy of water

As temperature increases, the $T\Delta S$ term grows in magnitude, progressively amplifying the entropic penalty associated with freezing. Below 273 K, this penalty is insufficient to offset the enthalpic driving force, and the solid phase remains favoured. At 273 K, the two contributions are exactly balanced ($\Delta G = 0$), marking the equilibrium melting point. Above this temperature, the entropic penalty outweighs the enthalpic driving force, rendering freezing non-spontaneous and the liquid phase thermodynamically favoured.

The Gibbs free energy equation provides a unified framework for evaluating reaction spontaneity by weighing the enthalpic and entropic contributions to the total free energy change. At elevated temperatures, the $T\Delta S$ term is amplified, increasing the thermodynamic reward for reactions that generate greater numbers of accessible microstates. This reflects the fact that at higher temperatures, the system possesses greater thermal energy to distribute across available microstates, making the free energy difference between high- and low-entropy states more pronounced. In this sense, the universe can be understood as evolving perpetually towards states of minimum free energy—that is, towards configurations with the greatest capacity for spontaneous change.

3.2 Free Energy & Equilibrium

Core Concept 3.5: Gibbs Free Energy & Equilibrium

We have established that a negative ΔG indicates a thermodynamically favorable reaction, while a positive ΔG indicates an unfavorable one. However, as seen in Section 2, a favorable reaction does not necessarily proceed to completion. Many reactions reach an equilibrium state in which both reactants and products coexist, regardless of the sign of ΔG under standard conditions. This implies a quantitative relationship between ΔG and the equilibrium constant K, given by the **Van't Hoff isotherm**.

$$\Delta G^\circ = -RT \ln K$$

This logarithmic relationship is consistent with Boltzmann's entropy expression $S = k_B \ln(W)$, where entropy scales logarithmically with the number of accessible microstates. Since the number of microstates available to a system grows exponentially with the number of molecules, and K reflects the ratio of product to reactant concentrations at equilibrium, a linear change in ΔG° corresponds to an exponential change in K.

Temperature plays a dual role in this relationship. It directly scales the magnitude of ΔG° and also shifts K through its dependence on ΔH, as described by the **Van't Hoff equation**, which describes how K changes with temperature.

$$\frac{d}{dt} \ln K_{eq} = \frac{\Delta H}{RT}$$

Through a simplier lense, this dependence can also be explained via the **Le Chatelier's principle** (*Core Concept 2.6*). The temperature at which $K = 1$ marks the point where neither the forward nor reverse reaction is thermodynamically preferred, corresponding to an equal ratio of reactants to products at equilibrium.

Under non-standard conditions, the free energy available to drive a reaction depends on the reaction quotient Q, which describes the current ratio of products to reactants. The free energy change under these conditions is given by:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

or equivalently:

$$\Delta G = -RT \ln \frac{K}{Q}$$

When $Q = K$, the system is at equilibrium and $\Delta G = 0$, meaning no further net work can be extracted from the reaction. When $Q \neq K$, the sign of ΔG indicates the direction in which the reaction will spontaneously proceed: a negative value favors the forward reaction, and a positive value favors the reverse.

4 Examples and Applications

4.1 Work

Core Concept 4.1: Free Energy & Work

The free energy released during a thermodynamic process can be harnessed to perform **work**, thereby **exerting influence on external systems**. Although endothermic reactions with sufficiently large positive entropy changes can be thermodynamically favorable and capable of performing work, exothermic reactions more commonly serve as the primary source of useful work. In many instances, exothermic processes are accompanied by entropy decreases, which reduces the magnitude of free energy available for driving external processes relative to the total heat released.

The practical significance of Gibbs free energy lies precisely in this capacity to quantify available work.

Revisiting the fundamental Gibbs free energy equation:

$$\Delta G = \Delta H - T\Delta S$$

This relationship reveals several key insights:

- When $\Delta H < 0$ (*exothermic*), the enthalpy change can be converted into work after accounting for the entropy cost imposed by the second law of thermodynamics.
- When $\Delta S < 0$ (*entropy decreasing*), the system must compensate for this entropy decrease to satisfy thermodynamic constraints—this compensation represents the *entropy cost*.
- When $\Delta S > 0$ (*entropy increasing*), the favorable entropy change itself contributes additional free energy that can be converted into work.

Core Concept 4.2: Work Example (*Rube Goldberg Machine*)

To illustrate how changes in entropy and enthalpy can be systematically converted into useful work, consider the coupled mechanical system depicted in Figure 11.

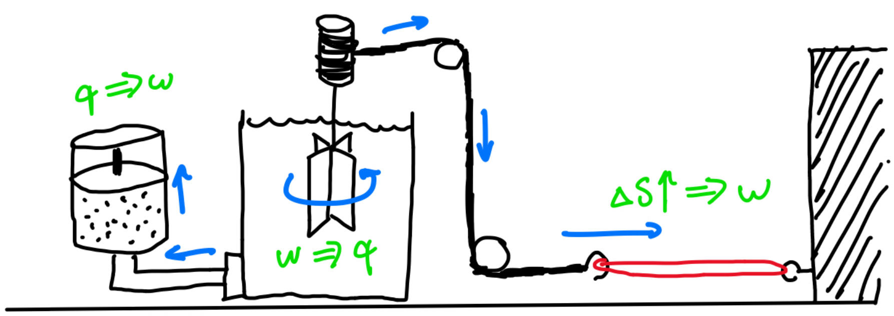

Figure 11: Entropy to work conversion

The system begins with a stretched rubber band in a low-entropy configuration. This constrained state has fewer accessible microstates compared to the relaxed configuration, where the polymer chains can explore a broader conformational range.

The system goes through the consequent chain reaction.

(a) **Entropy-to-Work Conversion:** As the rubber band spontaneously contracts toward its equilibrium state, the accompanying increase in entropy releases free energy. This free energy performs mechanical work by pulling the attached cable.

(b) **Work-to-Heat Conversion:** The mechanical work transmitted through the cable system rotates a paddle immersed in water. Frictional forces immediately dissipate this organized mechanical energy as heat into the water, increasing the thermal energy of the surroundings.

(c) **Heat-to-Work Conversion via Thermal Expansion:** The thermal energy is subsequently transferred to a gas confined within a piston chamber. As the gas absorbs heat, its entropy increases through volume expansion, performing work by displacing the piston.

A fundamental thermodynamic constraint emerges in this final conversion: **the conversion of heat to work is inherently irreversible and incomplete**. This limitation arises from the qualitative distinction between these energy forms. Heat represents disordered molecular motion—a high-entropy form of energy. Work, conversely, represents organized, directional motion—a low-entropy form. **Perfect heat-to-work conversion would violate the second law of thermodynamics by decreasing universal entropy**. Consequently, a portion of the thermal energy must be rejected to a lower-temperature reservoir, thereby generating sufficient entropy to satisfy thermodynamic requirements and compensate for the entropy decrease associated with converting random thermal motion into organized work. This heat-to-work conversion inefficiency is called the **Kelvin-Planck statement**.

Core Concept 4.3: Internal Energy

The **internal energy (E)** of a thermodynamic system represents the total kinetic and potential energy of all constituent particles. Changes in internal energy are governed by the first law of thermodynamics.

$$\Delta E = q + w$$

where q represents heat transferred and w represents work performed. Importantly, energy conservation dictates that any energy gained or lost by the system must be correspondingly lost or gained by the surroundings.

The relationship between heat input and internal energy change depends critically on the constraints imposed on the system. Consider two gas systems, each heated from an initial temperature T_{initial} to a final temperature T_{final}, with one maintained at constant volume and the other at constant pressure.

(a) **Constant Volume Process:** Under isochoric conditions, the system cannot perform expansion work ($w = 0$). Consequently, all heat transferred to the system increases its internal energy, manifesting entirely as a temperature rise.

(b) **Constant Pressure Process:** Under isobaric conditions, the system expands as temperature increases, performing pressure-volume work on the surroundings. To achieve the same temperature increase as the constant volume case, additional heat must be supplied to compensate for the energy transferred out of the system through expansion work.

This analysis demonstrates that achieving a given temperature increase under constant pressure conditions requires greater heat input than under constant volume conditions, with the additional energy accounting for the work performed during expansion.

4.2 Electrochemical Cells

Core Concept 4.4: Electrochemical Cells

An **electrochemical cell** is a device that converts chemical energy into electrical energy by harnessing the free energy released from spontaneous chemical reactions. The fundamental operating principle involves the spatial separation of a **reduction-oxidation** reaction, forcing e^-s to traverse an external circuit rather than transferring directly between species within a solution, thereby enabling the extraction of useful electrical work.

Core Concept 4.5: Electrode Potential

To understand electrochemical cells quantitatively, we must first establish the concept of electrode potential. **Electrode potential** quantifies the thermodynamic tendency of a metal electrode to either undergo oxidation (*releasing metal ions into solution*) or reduction (*depositing metal ions from solution onto the electrode surface*). *Note* *that these definitions of oxidation and reduction are specifically within the context of electrochemical cells.*

A high (*positive*) electrode potential indicates a strong thermodynamic driving force for reduction, wherein metal ions in solution are deposited as neutral metal atoms on the electrode surface.

$$M^+(aq) + e^- \longrightarrow M(s)$$

Conversely, a low (*negative*) electrode potential indicates a propensity for oxidation, wherein neutral metal atoms dissolve into solution as cations:

$$M(s) \longrightarrow M^+(aq) + e^-$$

Since absolute electrode potentials cannot be measured in isolation, all electrode potentials are reported relative to a universal reference standard. The **Standard Hydrogen Electrode (SHE)** serves this purpose, arbitrarily assigned a potential of 0.00 V under standard conditions (*298 K, 1 atm, 1 M concentration*). Table 3 presents standard reduction potentials for common half-reactions.

Potential	Reduction Half-Reaction
−3.05 V	$Li^+(aq) + e^- \longrightarrow Li(s)$
−2.71 V	$Na^+(aq) + e^- \longrightarrow Na(s)$
−2.37 V	$Mg^{2+}(aq) + 2\,e^- \longrightarrow Mg(s)$
−1.66 V	$Al^{3+}(aq) + 3\,e^- \longrightarrow Al(s)$
−0.76 V	$Zn^{2+}(aq) + 2\,e^- \longrightarrow Zn(s)$
−0.28 V	$Ni^{2+}(aq) + 2\,e^- \longrightarrow Ni(s)$
0.00 V	$2\,H^+(aq) + 2\,e^- \longrightarrow H_2(g)$
+0.34 V	$Cu^{2+}(aq) + 2\,e^- \longrightarrow Cu(s)$
+0.80 V	$Ag^+(aq) + e^- \longrightarrow Ag(s)$
+1.06 V	$Br_2(l) + 2\,e^- \longrightarrow 2\,Br^-(aq)$
+1.23 V	$O_2(g) + 4\,H^+(aq) + 4\,e^- \longrightarrow 2\,H_2O(l)$
+1.36 V	$Cl_2(g) + 2\,e^- \longrightarrow 2\,Cl^-(aq)$
+2.87 V	$F_2(g) + 2\,e^- \longrightarrow 2\,F^-(aq)$

Table 3: Standard reduction potentials of common half-reactions

Electrode potentials are expressed in volts (V), equivalent to joules per coulomb (J/C), reflecting the energy per unit charge that e^-s acquire as they spontaneously flow from the electrode of lower reduction potential to that of higher reduction potential.

Core Concept 4.6: Galvanic Cell

Consider the spontaneous reaction that occurs when zinc metal is immersed in a copper(II) ion solution.

$$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$$

This thermodynamically favorable process ($\Delta G < 0$) releases free energy that would otherwise escape as heat. A **galvanic cell** is an electrochemical device designed to capture this free energy by spatially separating the oxidation and reduction half-reactions, forcing e^-s through an circuit where they can perform electrical work.

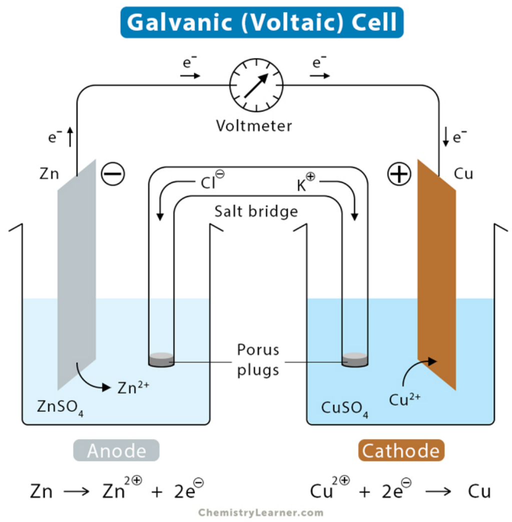

Figure 12: Galvanic cell diagram

In the galvanic cell shown in Figure 12, the electrode of lower reduction potential serves as the **anode** (*site of oxidation*) and the electrode of higher reduction potential serves as the **cathode** (*site of reduction*)—zinc and copper, respectively, in the above schematic. The physical separation of the two half-cell compartments necessitates a salt bridge to maintain electroneutrality. Without the salt bridge, charge imbalance would develop from the accumulation of Zn^{2+} at the anode and depletion of Cu^{2+} at the cathode, creating an electrostatic barrier that would arrest the reaction. The salt bridge permits compensatory ion migration, sustaining continuous e^- flow through the external circuit until the limiting reactant is fully consumed.

The relationship between cell potential and Gibbs free energy is given by the following equation.

$$\Delta G^\circ = -nFE^\circ$$

where n is the moles of e^-s transferred, F is the Faraday constant ($96485\,\mathrm{C\,mol^{-1}}$), and E° is the standard cell potential, defined as the difference between the cathode and anode reduction potentials.

For the zinc-copper system (*Figure 12*):

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.34\,\mathrm{V} - (-0.76\,\mathrm{V}) = 1.10\,\mathrm{V}$$

With $n = 2$ mol electrons transferred per mole of reaction:

$$\Delta G^\circ = -(2\,\mathrm{mol})(96485\,\mathrm{C\,mol^{-1}})(1.10\,\mathrm{V}) = -212.3\,\mathrm{kJ}$$

This value represents the maximum electrical work extractable **per mole** of reaction under standard conditions—equivalently, per mole of zinc oxidized when zinc serves as the limiting reactant.

Core Concept 4.7: Concentration Cells

A **concentration cell** is a specialized electrochemical system in which identical electrode materials are employed at both the anode and cathode, with the sole thermodynamic driving force arising from the difference in electrolyte concentration between the two half-cell compartments.

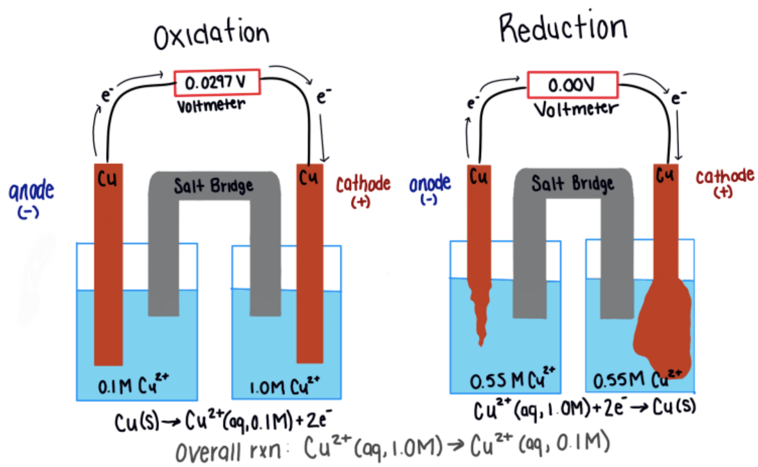

Figure 13: Concentration cell diagram

The net redox reaction for a copper concentration cell is:

$$\mathrm{Cu_{anode}(s) + Cu_{cathode}{}^{2+}(aq) \rightleftharpoons Cu_{anode}{}^{2+}(aq) + Cu_{cathode}(s)}$$

At equilibrium, when the concentrations of both compartments are equal, $K = 1$. Under non-equilibrium conditions, however:

$$Q = \frac{[\mathrm{Cu_{anode}}^{2+}]}{[\mathrm{Cu_{cathode}}^{2+}]} < 1$$

Since $Q < K$, the forward reaction is thermodynamically favored, driving oxidation at the dilute anode and reduction at the concentrated cathode **until the two compartments reach equal concentration**. This concentration gradient serves as the sole source of free energy driving e^- transport through the external circuit.

The cell potential under non-standard conditions is derived by combining the relevant thermodynamic relationships.

$$(\textit{Standard Conditions}) \quad \Delta G^\circ = -nFE^\circ = -RT\ln(K)$$

$$(\textit{Non-Standard Conditions}) \quad \Delta G = -nFE$$

$$\Delta G = \Delta G^\circ + RT\ln(Q)$$

$$-nFE = -nFE^\circ + RT\ln(Q)$$

$$(\textit{Nernst Equation}) \quad E = E^\circ - \frac{RT}{nF}\ln(Q)$$

For concentration cells, $E^\circ = 0$ since the same half-reaction occurs at both electrodes. Substituting standard temperature (*298 K*) and converting to base-10 logarithms yields the **simplified Nernst equation.**

$$E = -\frac{0.0592}{n}\log\left(\frac{[\mathrm{M}^{n+}_{\text{anode}}]}{[\mathrm{M}^{n+}_{\text{cathode}}]}\right)$$

For the copper concentration cell with $[\mathrm{Cu_{anode}}^{2+}] = 0.1\,\mathrm{M}$ and $[\mathrm{Cu_{cathode}}^{2+}] = 1.0\,\mathrm{M}$, the cell potential is $E = 0.0296\,\mathrm{V}$. This characteristically low potential, inherent to concentration-driven cells, limits their practical utility as energy sources relative to conventional galvanic cells employing dissimilar electrode materials.

Core Concept 4.8: Hydrogen Fuel Cell

A **hydrogen fuel cell** represents a distinct class of electrochemical device in which gaseous **hydrogen** serves as the anode reactant and gaseous **oxygen** serves as the cathode reactant, harnessing the electron transfer energy accompanying the thermodynamically favorable formation of water.

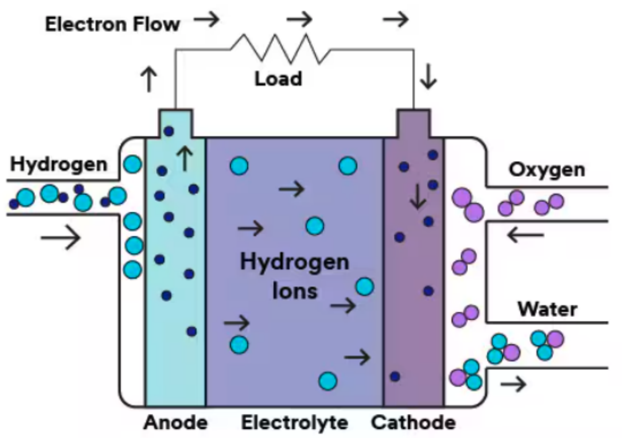

Figure 14: Hydrogen fuel cell diagram

The overall cell reaction is:

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$$

Decomposing this into constituent half-reactions, hydrogen oxidation occurs at the anode:

$$H_2(g) \longrightarrow 2\,H^+(aq) + 2\,e^-$$

while oxygen reduction occurs at the cathode:

$$\frac{1}{2}O_2(g) + 2\,H^+(aq) + 2\,e^- \longrightarrow H_2O(l)$$

Applying the same thermodynamic formalism established for galvanic cells (*Core Concept 4.6*), the standard cell potential is:

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 1.23\,\text{V} - 0.00\,\text{V} = 1.23\,\text{V}$$

$$\Delta G^\circ = -(2\,\text{mol})(96485\,\text{C mol}^{-1})(1.23\,\text{V}) = -237.4\,\text{kJ}$$

This free energy value, corresponding to the oxidation of one mole of hydrogen gas, reflects the substantial energy density of hydrogen as a fuel and *underscores the electrochemical efficiency that makes hydrogen fuel cells a compelling candidate for clean energy applications*.